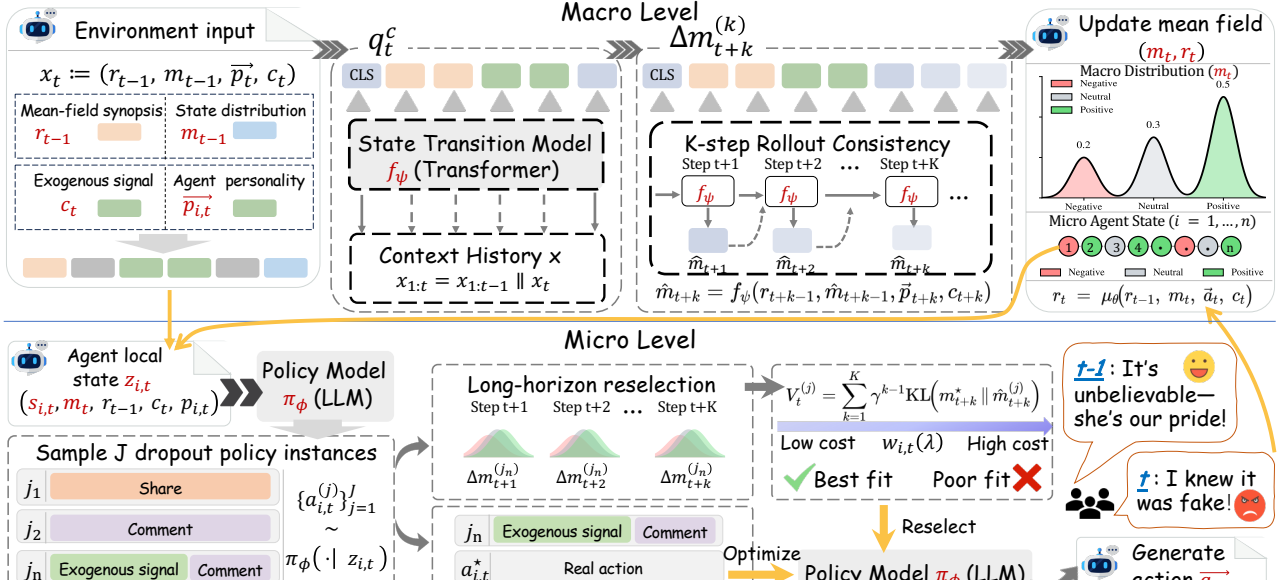
Macro Level
Environment input
$x_t := (r_{t-1}, m_{t-1}, \vec{p}_t, c_t)$
Mean-field synopsis
r_{t-1}
State distribution
m_{t-1}
Exogenous signal
c_t
Agent personality
$\vec{p}_{i,t}$
q_t^c
CLS
State Transition Model
f_ψ (Transformer)
Context History x
$x_{1:t} = x_{1:t-1} \| x_t$
$\Delta m_{t+k}^{(k)}$
CLS
K-step Rollout Consistency
Step t+1
Step t+2
...
Step t+K
f_ψ
$\hat{m}_{t+1}$
$\hat{m}_{t+2}$
$\hat{m}_{t+k}$
$\hat{m}_{t+k} = f_\psi(r_{t+k-1}, \hat{m}_{t+k-1}, \vec{p}_{t+k}, c_{t+k})$
Update mean field
(m_t, r_t)
Macro Distribution (m_t)
Negative
Neutral
Positive
0.2
0.3
0.5
Micro Agent State ($i = 1, \ldots, n$)
1
2
3
4
n
Negative
Neutral
Positive
$r_t = \mu_\theta(r_{t-1}, m_t, \vec{a}_t, c_t)$
Micro Level
Agent local state $z_{i,t}$
$(s_{i,t}, m_t, r_{t-1}, c_t, p_{i,t})$
Policy Model
π_ϕ (LLM)
Sample J dropout policy instances
j_1 Share
j_2 Comment
j_n Exogenous signal Comment
$\{a_{i,t}^{(j)}\}_{j=1}^J \sim \pi_\phi(\cdot \mid z_{i,t})$
Long-horizon reselection
Step t+1
Step t+2
...
Step t+K
$\Delta m_{t+1}^{(j_n)}$
$\Delta m_{t+2}^{(j_n)}$
$\Delta m_{t+k}^{(j_n)}$
$V_t^{(j)} = \sum_{k=1}^{K} \gamma^{k-1} \mathrm{KL}\left(m_{t+k}^\star \| \hat{m}_{t+k}^{(j)}\right)$
Low cost
$w_{i,t}(\lambda)$
High cost
Best fit
Poor fit
j_n Exogenous signal Comment
$a_{i,t}^\star$ Real action
Optimize
Reselect
Policy Model π_ϕ (LLM)
Generate action $\vec{a}_{i,t}$
t-1: It's unbelievable—she's our pride!
t: I knew it was fake!